RIS-Assisted Flexible Space Shift Keying for Wireless Communication System

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***Abstract*—In this letter, we propose a novel communication scheme that combines reconfigurable intelligent surface with transmitted flexible space shift keying (RIS-TFSSK) for future wireless communication. In RIS-TFSSK scheme, the incoming bits are divided into two parts, where the first part is used to determine the number of active antennas and the second part is used to determine the transmit antenna combinations. Besides, an antenna selection method is proposed to select the candidate antenna combinations with largest channel gains, which further improves the bit error rate (BER) performance of RIS-TFSSK scheme. By comparing with the RIS-aided transmitted space shift keying and RIS-aided transmitted generalized space shift keying, the theoretical and simulation results show that the proposed scheme has better BER performance.**

***Index Terms*—Reconfigurable intelligent surface, flexible space shift keying, antenna selection, bit error ratio.**

I. Introduction

INDEX modulation (IM) has been widely studied in recent years, and it has become an attractive research direction for the next generation wireless communication due to its advantages of low complexity, high spectral efficiency, and high energy efficiency [1]. In IM, one popular paradigm is spatial modulation (SM) [2], [3] which uses the antenna index to carry additional bits and transmit traditional phase shift keying (PSK) and quadrature amplitude modulation (QAM) by the selected active antenna. Space shift keying (SSK) and generalized space shift keying (GSSK) as simplified forms of SM are proposed in [4] and [5]. In [6], flexible spatial modulation (FSM) as a high spectral efficiency scheme provides more possibilities to solve the spectral efficiency problem.

On the other hand, sixth-generation (6G) wireless communication requires new communication paradigms to achieve ultra-high user throughput as well as high spectral and energy efficiency, which puts the focus on the changes at the physical layer. In this way, the emergence of reconfigurable intelligent surface (RIS) provides opportunities to solve the aforementioned issue and RIS may become one candidate technology for 6G communications [7]. The RIS improves communication quality and signal coverage by intentionally manipulating the characteristics of electromagnetic waves in the propagating environment through the controller. Each reflector can independently adjust the phase of the incident wave reflected to the receiver, thereby creating a more efficient wireless channel. Considering the above two technologies, i.e., IM and RIS, the combination of RIS and IM has become a new popular research trend. The RIS assisted IM schemes were proposed in [8], [9], [10], [11], [12], which provide the development prospect of combining the RIS and IM. Admittedly, the combination of RIS and IM improves the quality of communication, but the IM in these combination technologies usually suffers the low spectral efficiency problem. Motivated by FSM and the SSK, a FSSK scheme should be taken into account as the IM to combine with RIS to boost the overall system performance. To the best of our knowledge, no existing studies consider the combination of FSSK and RIS to propose a high spectral efficiency and low complexity transmission scheme.

To fill this blank, a reconfigurable intelligent surface with transmitted flexible space shift keying (RIS-TFSSK) is proposed in this letter. In the proposed scheme, the incoming bits are divided into two parts, where the first part is used to determine the number of active antennas and the second part is used to determine the selected antenna combinations for transmission. In this way, the transmitter not only can carry additional information but also can be used flexibly. When the number of transmit antennas increases, the system performance of data rate becomes more obvious. Then, based on the RIS-TFSSK system model, the theoretical derivation of bit error rate (BER) and system complexity analysis are given. The BER performance of the proposed scheme is demonstrated by Monte Carlo simulation and RIS-TFSSK is further compared with RIS assisted transmitted space shift keying (RIS-TSSK) scheme [13] and RIS assisted transmitted generalized space shift keying (RIS-TGSSK) benchmark scheme under the same spectral efficiency. In addition, the potential of RIS-TFSSK scheme is evaluated under different setting of antenna combination. Moreover, an antenna selection algorithm based on largest channel gain is designed according to the channel estimation method proposed in [14], which provides better BER performance.[1]

II. System Model

In this letter, a RIS assisted TFSSK system with N_t transmit antennas and single receive antenna is considered in Fig. 1, where the direct link between source and destination is blocked by a barrier. In order to realize the communication, the system

Manuscript received 29 May 2023; revised 28 June 2023; accepted 30 June 2023. Date of publication 11 July 2023; date of current version 9 October 2023. This work was supported by the Science and Technology Development Fund, Macau SAR (0108/2020/A3) and (0005/2021/ITP). The associate editor coordinating the review of this article and approving it for publication was B. Makki. *(Corresponding author: Yuyang Peng.)*

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Digital Object Identifier 10.1109/LWC.2023.3293837

[1] *Notation:* Bold, lowercase and capital letters are used for column vectors and matrices, respectively. The real part of a complex variable is denoted by $\Re\{\cdot\}$. $M_X(\cdot)$ stands for the moment generating function (MGF) of a random variable (RV) X. $Q(\cdot)$ is the Gaussian Q-function.

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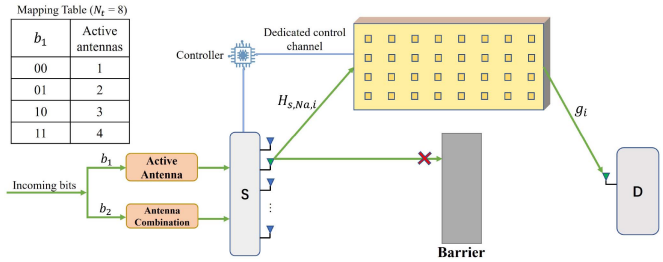

Fig. 1. RIS-assisted TFSSK system model.

is established with the aid of the RIS, which consists of N low-cost passive reflection elements and can adjust the phase of the incident wave based on channel state information (CSI). The channel from the transmitter to the RIS and the RIS to the receiver are denoted as $\mathrm{H} \in \mathbb{C}^{N_t \times N}$ and $\mathrm{G} \in \mathbb{C}^{N \times 1}$, respectively. It is also assumed that the CSI of these two channels are known in RIS via a communication software [8], [9], [10], [11]. Note that if the perfect CSI is not known, alternative imperfect CSI can be used as the other choice via channel estimation methods [15], which is out of the scope of our work and will not be discussed in this letter.

In the RIS-TFSSK system model, the incoming bits are divided into two main subgroups, i.e., b_1 and b_2. The first b_1 bit group represents the number of active antennas out of N_t transmit antennas and denoted as N_a, where $1 \leq N_a \leq N_t/2$. There is a mapping table for active antennas when $N_t = 8$ shown in Mapping Table as an example. After N_a is determined, the number of all possible antenna combinations can be calculated as $\mathrm{J} = \binom{N_t}{N_a}$ and the full set of antenna combinations is denoted as $\mathcal{A}$. At this point, S out of J transmit antenna combinations are selected to map for the incoming bits b_2.[2] Therefore, the length of b_1 and b_2 are determined by $\log_2(N_t/2)$ and $\log_2(\mathrm{S})$, respectively, and the spectral efficiency (bits/s/Hz) can be calculated by $r = \log_2(N_t/2) + \log_2 \mathrm{S}$.

A. Transmission

In this subsection, the transmission of RIS-TFSSK scheme is given. The wireless fading channel from the m-th transmit antenna combination to the i-th reflection element and the i-th reflection element to the receiver are given by $\mathrm{H}_{m,i} = \alpha_{m,i} e^{-j\theta_{m,i}}$ and $G_i = \beta_i e^{-j\psi_i}$, respectively, where $m \in \{1, 2, \ldots, N_t\}$ and $i \in \{1, 2, \ldots, N\}$. The $\alpha_{m,i}$ and β_i represent the Rayleigh factors under the flat Rayleigh fading channels, which are assumed to be an independent and identically distributed complex Gaussian random variable with $\mathcal{CN}(0,1)$. Since N_a is determined, a new combination set is expressed as $K_{N_a} = \{\hat{\mathbf{H}}_{1,N_a,i}, \hat{\mathbf{H}}_{2,N_a,i}, \ldots, \hat{\mathbf{H}}_{j,N_a,i}, \ldots, \hat{\mathbf{H}}_{\mathrm{J,N_a,i}}\}$, which includes all antenna combinations with N_a active antennas, and the construction of j-th element of K_{N_a} can be calculated from:

$$\hat{\mathbf{H}}_{j,N_a,i} = \frac{1}{\sqrt{N_a}} \sum_{m \in \mathcal{A}(j)} \mathbf{H}_{m,i}. \tag{1}$$

[2]Here, it is worth noting that the number of transmit antennas N_t and the number of selected antenna combinations S must be power of two ($\mathrm{S} \leq N_t$).

TABLE I
MAPPING FOR SELECTED ANTENNA COMBINATIONS WITH $N_t = 8$

b_2	Selected combinations	Antenna combination
000	$\hat{\mathrm{H}}_{1,2}$	$(1, 2)$
001	$\hat{\mathrm{H}}_{2,2}$	$(2, 8)$
010	$\hat{\mathrm{H}}_{3,2}$	$(3, 4)$
011	$\hat{\mathrm{H}}_{4,2}$	$(1, 6)$
100	$\hat{\mathrm{H}}_{5,2}$	$(4, 7)$
101	$\hat{\mathrm{H}}_{6,2}$	$(2, 5)$
110	$\hat{\mathrm{H}}_{7,2}$	$(3, 6)$
111	$\hat{\mathrm{H}}_{8,2}$	$(5, 8)$

In each time slot, S out of J transmit antenna combinations from K_{N_a} are selected, and the corresponding channel is denoted as $\hat{\mathbf{H}}_{s,N_a,i} = \alpha_{s,N_a,i} e^{-j\theta_{s,i}}$, with $\alpha_{s,N_a,i}$ and $e^{-j\theta_{s,i}}$ being the Rayleigh factor and the channel phase of s-th selected transmit combination, respectively, where s is the index of determined combination $s \in \{1, 2, \ldots, \mathrm{S}\}$. The new candidate antenna combinations can be denoted as $\hat{K}_{N_a} = \{\hat{\mathbf{H}}_{1,N_a,i}, \hat{\mathbf{H}}_{2,N_a,i}, \ldots, \hat{\mathbf{H}}_{s,N_a,i}, \ldots, \hat{\mathbf{H}}_{\mathrm{S},N_a,i}\}$. A sample mapping table for the selected combinations under b_2 with $N_t = 8$, $\mathrm{S} = 8$, and $N_a = 2$ determined by $b_1 = 01$ is shown in Table I with the spectral efficiency is 5 bits/s/Hz. Therefore, the received signal of the s-th selected combination is expressed as:

$$y = \sqrt{E}\left(\sum_{i=1}^{N} \alpha_{s,N_a,i} \beta_i e^{j(\phi_{s,i} - \theta_{s,i} - \psi_i)}\right) + n, \tag{2}$$

where E is the transmitted signal energy, n is the additive white Gaussian noise term with zero-mean and variance N_0, $\phi_{s,i}$ represents the adjusted phase for the i-th reflection element of the RIS, determined by a communication software from the s-th selected antenna combination. The instantaneous signal-to-noise ratio (SNR) at receiver is calculated as:

$$\mathrm{SNR} = \frac{E\left|\sum_{i=1}^{N} \alpha_{s,N_a,i} \beta_i e^{j(\phi_{s,i} - \theta_{s,i} - \psi_i)}\right|^2}{N_0}, \tag{3}$$

Obviously, the maximum SNR can be obtained when $\phi_i = \theta_{s,i} + \psi_i$, which can be expressed as:

$$\max\{\mathrm{SNR}\} = \frac{E\left|\sum_{i=1}^{N} \alpha_{s,N_a,i} \beta_i\right|^2}{N_0}. \tag{4}$$

B. Detection

Assuming that the CSI is perfectly known at the receiver, the ML detector is considered for RIS-TFSSK to jointly detect the number of active antennas and the selected transmit antenna or antenna combination by traversing all possibilities, which is given as:

$$[\hat{N}_a, \hat{s}] = \arg\min_{N_a, s} \left\{|y - \tilde{y}|^2\right\}, \tag{5}$$

where $\hat{N}_a$ and $\hat{s}$ stand for the estimated number of active antennas and the estimated antenna combination, respectively,

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y represents the received signal of selected transmit antenna or antenna combination, and we define:

$$\tilde{y} = \sqrt{E}\left(\sum_{i=1}^{N} \alpha_{\hat{s},\hat{N}_a,i}\beta_i e^{j(\phi_{\hat{s},i}-\theta_{\hat{s},i}-\psi_i)}\right), \tag{6}$$

which represents the received signal is estimated and conveyed by $\hat{N}_a$ active antennas and $\hat{s}$-th antenna combination.

III. Performance Analysis

In this section, the analytical derivation of the RIS-TFSSK scheme in terms of the theoretical BER over Rayleigh fading channels is proposed. Suppose that the number of active antennas and the selected antenna combination are N_a and s, which is denoted as L, and its erroneous detection are $\hat{N}_a$ and $\hat{s}$, which is denoted as P. The conditional pairwise error probability (PEP) can be obtained as:

$$\begin{aligned}
&\Pr\{L \to P \mid \mathbf{H}\} \\
&= \Pr\left\{\left|y-\sqrt{E}H_L\right|^2 > \left|y-\sqrt{E}H_P\right|^2\right\} \\
&= \Pr(\left\{\left|y-\sqrt{E}H_L\right|^2 - \left|y-\sqrt{E}H_P\right|^2\right\} > 0) \\
&= \Pr\left\{-E|H_L-H_P|^2 - 2\Re\left\{\sqrt{E}(H_L-H_P)n^*\right\} > 0\right\} \quad (7)
\end{aligned}$$

where

$$\begin{aligned}
H_L &= \sum_{i=1}^{N} \alpha_{s,N_a,i}\beta_i, \\
H_P &= \sum_{i=1}^{N} \alpha_{\hat{s},\hat{N}_a,i}\beta_i e^{j(\phi_{\hat{s},i}-\theta_{\hat{s},i}-\psi_i)} = \sum_{i=1}^{N} \alpha_{\hat{s},\hat{N}_a,i}\beta_i. \quad (8)
\end{aligned}$$

It is apparent that if the CSI is known at the RIS, the phase $\phi_{\hat{s},i}$ can be adjusted to the optimal phase even it transmits using the estimated antenna combination, i.e., $\phi_{\hat{s},i} = \theta_{\hat{s},i} + \psi_i$. To facilitate subsequent calculations, we define this probability event as $\Pr\{F > 0\}$ with F being a complex Gaussian RV, which follows the distribution with mean $\mu_F = -E|H_L - H_P|^2$ and variance $\sigma_F^2 = 2EN_0|H_L - H_P|^2$. In (7), n^* is the complex conjugate value of n. Consequently, from $P\{F > 0\} = Q(-\mu_F/\sigma_F)$, the conditional PEP can be expressed as:

$$\Pr(L \to P \mid H) = Q\left(\frac{E|H_L-H_P|^2}{\sqrt{2EN_0|H_L-H_P|^2}}\right). \tag{9}$$

Additionally, by denoting $D = H_L - H_P$, it can be extended as:

$$D = \sum_{i=1}^{N} \beta_i\left(\alpha_{s,N_a,i} - \alpha_{\hat{s},\hat{N}_a,i}\right), \tag{10}$$

which follows the Gaussian distribution with zero mean and variance of σ_D^2. Since RIS is usually equipped with a large number of reflection elements, according to the central limit theorem (CLT), σ_D^2 is calculated as $N(2-\pi/2)$.[3]

[3]If X and Y are independent RVs, which follow the same distribution, Var{XY} = Var{X}Var{Y}+Var{X}E^2{Y}+E^2{X}Var{Y}, Var{X+Y} = Var{X}+Var{Y} and the variance of $(\alpha_{s,N_a,i} - \alpha_{\hat{s},\hat{N}_a,i})$ and β_i can be obtained as $Var\{\cdot\} = (2-\pi/2)$ and $Var\{\beta_i\}] = 4-\pi/4$, respectively.

To calculate the average PEP (APEP), we define $\nu = |D|^2$, which follows a central chi-square distribution with one degree of freedom, and the APEP is obtained as:

$$\begin{aligned}
\overline{P}_e &= \int_0^{\infty} Q\left(\sqrt{\frac{E\nu}{2N_0}}\right) f_\nu(\nu)\, d\nu \\
&= \frac{1}{\pi}\int_0^{\frac{\pi}{2}} M_\nu\left(-\frac{E}{4N_0\sin^2\eta}\right) d\eta, \quad (11)
\end{aligned}$$

where $M_\nu(t) = (1-2\sigma_D^2 t)^{-1/2}$ by defining $t = -\frac{E}{4N_0\sin^2\eta}$ in [13], and then substitute it in (11), and the APEP can be expressed as:

$$\overline{P}_e = \frac{1}{\pi}\int_0^{\frac{\pi}{2}} \left(\frac{1}{\sqrt{1+\left(2-\frac{\pi}{2}\right)\frac{NE}{2N_0\sin^2\eta}}}\right) d\eta. \tag{12}$$

IV. Antenna Selection and Complexity Analysis

In this section, we explain the working principle of RIS-TFSSK scheme with antenna selection (RIS-AS-TFSSK) and the system complexity is also calculated.

A. RIS-AS-TFSSK

Since the available antenna combinations are more than the selected antenna combinations when multiple antennas are selected, this causes a certain waste of resources. In order to prevent this phenomenon and obtain better BER performance, we can select the antenna combinations with largest channel gains. After N_a is determined, the antenna combination set K_{N_a} is constructed by (1) and the channel coefficients from K_{N_a} are sorted with respect to their channel gains, which can be expressed as:

$$\left\|\hat{\mathbf{H}}_{1,N_a,i}\right\|^2 > \left\|\hat{\mathbf{H}}_{2,N_a,i}\right\|^2 > \left\|\hat{\mathbf{H}}_{3,N_a,i}\right\|^2 > \cdots > \left\|\hat{\mathbf{H}}_{\mathrm{J},N_a,i}\right\|^2. \tag{13}$$

Then, S out of J transmit antenna combinations with large channel gains are selected as candidate transmit combinations, which have a one-to-one mapping relationship with b_2 bit group.[4] In order to understand the process of antenna selection more clearly, an example is shown below for RIS-AS-TFSSK scheme with $N_t = 8$ and S = 4.

Example: In this case, $N_t = 8$ and $1 \le N_a \le 4$ according to the incoming bits. The data rate of this case is 4 bits/s/Hz. Assume that the incoming bits are 0110, which are separated into subgroups as $b_1 = 01$ and $b_2 = 10$. According to Mapping Table, the number of active antennas is determined as $N_a = 2$ when $b_1 = 01$. After N_a is determined, $\hat{\mathrm{H}}_{s,N_a} = \hat{\mathrm{H}}_{3,2}$ is determined when $b_2 = 10$, which can be seen from Table II. In this way, the third and sixth transmit antenna are selected to transmit information.

[4]For this purpose, the transmitter needs to know the CSI of the transmitting channel. We can use the unilateral channel estimation method proposed in [14] to estimate the whole channel by controlling the phase shift change of each reflection elements, and transmit the CSI to the transmitter through the controller.

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TABLE II
MAPPING FOR SELECTED ANTENNA COMBINATIONS WITH ANTENNA SELECTION WITH $N_t = 8$ AND S = 4

b_2	Selected combinations	Antenna combination
00	$\hat{\mathrm{H}}_{1,2}$	$(1,7)$
01	$\hat{\mathrm{H}}_{2,2}$	$(2,5)$
10	$\hat{\mathrm{H}}_{3,2}$	$(3,6)$
11	$\hat{\mathrm{H}}_{4,2}$	$(4,5)$

TABLE III
DETECTION COMPLEXITY ANALYSIS SUMMARY

Scheme	Complexity
RIS-TSSK	$(4N+2)N_t$
RIS-TGSSK	$\left(\left(3+\left(\frac{N_t}{2}+1\right)\right)N+2\right)2^m$
RIS-TFSSK	$\left(\left(3+\left(\frac{N_t}{4}+\frac{1}{2}\right)\right)N+2\right)2^m$

B. Complexity Analysis

The complexity of the antenna selection is defined as the number of complex multiplications (CMs) and complex additions (CAs) required to order the square of the channel vector norm. Considering this procedure and following (13), the average complexity of antenna selection is expressed as:

$$C_{\mathrm{Antenna-Selection}} = \frac{\sum_{N_a=1}^{N_t/2}\binom{N_t}{N_a}[(NN_a+2)]}{N_t/2}, \quad (14)$$

where $1 \leq N_a \leq N_t/2$. The receiver complexity of the RIS-TFSSK scheme is defined as the number of CMs and CAs required for ML detector in (5). The complexity of ML detection is given by:

$$C_{\mathrm{ML,RIS-TFSSK}} = \left(\left(3+\left(\frac{\frac{N_t}{2}+1}{2}\right)\right)N+2\right)2^r. \quad (15)$$

where $(\frac{\frac{N_t}{2}+1}{2})$ is the expected value of N_a.

The results of the complexity analysis are summarized in Table III, which also includes RIS-TSSK and RIS-TGSSK schemes. It is clear that RIS-TSSK scheme has lowest complexity, but compared with other schemes, it also requires the largest number of antennas to achieve the same spectral efficiency. RIS-TFSSK scheme has the second lowest complexity, but it has the highest spectral efficiency under the condition of the same antennas, which realizes the good tradeoff between complexity and spectral efficiency. Note that if the complexity is the main design purpose, we can also apply the sub-surface method to further reduce the complexity at the cost of losing some BER performance, which will not be discussed here in detail.

V. SIMULATION RESULTS

In this section, we analyze the BER performance of the proposed RIS-TFSSK scheme and RIS-AS-TFSSK scheme based on Monte Carlo simulation. In the simulation, the SNR value is considered as E/N_0 and under each SNR value 2×10^5 bits are sent to ensure the accuracy of the experimental results. By comparing with RIS-TSSK scheme [13] and RIS-TGSSK benchmark scheme, significant BER performance of the proposed scheme is demonstrated.

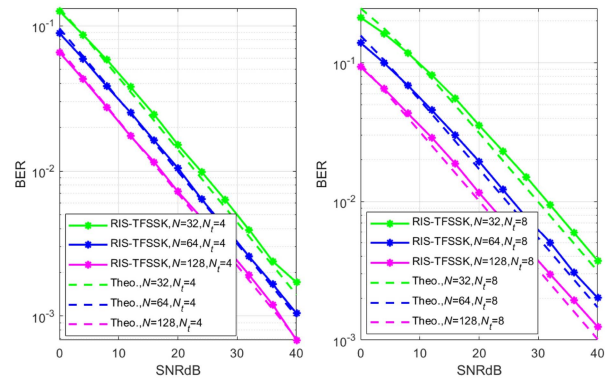

Fig. 2. The BER performance of RIS-TFSSK system with $N_t = 4$ and $N_t = 8$ under various N.

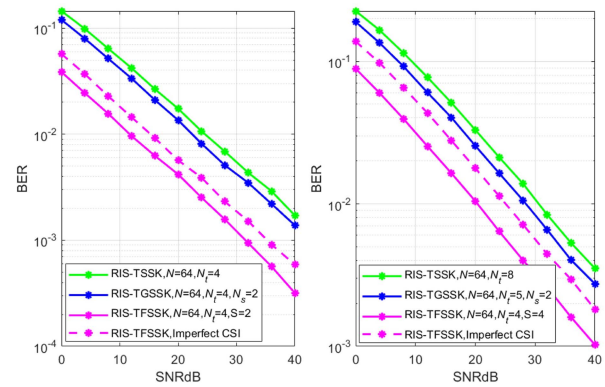

Fig. 3. Comparisons of BER performance in RIS-TFSSK with RIS-TFSSK with imperfect CSI, RIS-TSSK and RIS-TGSSK for 2 and 3 bits/s/Hz with $N = 64$.

Firstly, the effect of the number of reflection elements is investigated for the proposed RIS-TFSSK scheme in terms of BER through analytical derivation and simulation, which is shown in Fig. 2. It can be seen from the figure that for $N_t = 4$ and $N_t = 8$, doubling the number of reflective elements N allows the proposed scheme to achieve SNR gains of 3 dB and 4 dB under the same BER performance. In addition, the theoretical and simulation results of the RIS-TFSSK scheme match each other.

Fig. 3 shows the BER performance of various RIS-aided transmitted index modulation schemes including the proposed scheme with perfect and imperfect CSI, where we use $N = 64$ and imperfect CSI with variance of estimated error $\sigma^2 = 0.5$ [10], [15]. It can be seen from the figure that BER performance decreases with the increase of S. The reason is that increasing S means that more transmission combinations are selected so that the complexity of detection is increased. It is investigated that the RIS-TFSSK scheme has the worst BER performance under the imperfect CSI, which leads to 4 dB and 7 dB gains loss for 2 and 3 bits/s/Hz, respectively. The RIS-TFSSK scheme with S = 2 has approximately 12 dB and 10 dB gains at the BER of 10^{-2} with respect to RIS-TSSK scheme and RIS-TGSSK scheme, respectively, under the 2 bits/s/Hz. When S = 4, the BER performance of RIS-TFSSK scheme decreases, but the spectral efficiency increases to 3

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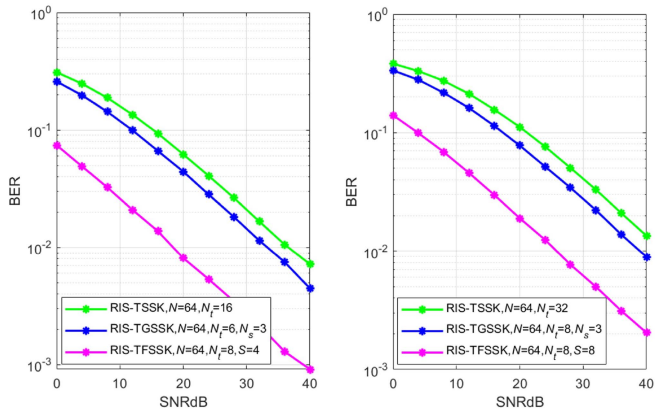

Fig. 4. Comparison of BER performance among RIS-TFSSK, RIS-TSSK and RIS-TGSSK for 4 and 5 bits/s/Hz with $N = 64$.

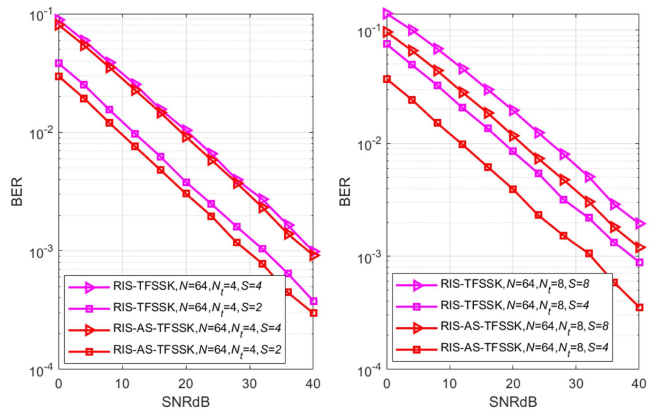

Fig. 5. The BER performance of RIS-AS-TFSSK and RIS-TFSSK with $N_t = 4$ and $N_t = 8$.

bits/s/Hz. To achieve the same spectral efficiency, RIS-TSSK scheme requires eight transmit antennas, while RIS-TGSSK scheme requires five antennas and two activated antennas in each time slot. At this point, the RIS-TFSSK scheme has about 10 dB and 8 dB gains at the BER of 10^{-2} compared with RIS-TSSK scheme and RIS-TGSSK scheme, respectively.

In Fig. 4, we extend our analysis to $N_t = 8$. In this case, the BER performance of RIS-TFSSK is considered when S $= 4$ and S $= 8$ under 4 and 5 bits/s/Hz. When the spectral efficiency is 4 bits/s/Hz, the RIS-TFSSK scheme brings 15 dB and 13 dB gains at the BER of 10^{-2} compared to RIS-TSSK scheme with $N_t = 16$ and RIS-TGSSK scheme with $N_t = 6$ and $N_s = 3$ (the number of selected antennas), respectively. In addition, we select S $= 8$, the spectral efficiency is increased to 5 bits/s/Hz. In this setting, RIS-TFSSK can achieve about 14 dB and 12 dB gains, respectively, under the same BER.

To illustrate the BER performance of RIS-AS-TFSSK scheme, we compare it with RIS-TFSSK scheme at $N_t = 4$ and $N_t = 8$, which can be seen in Fig. 5. In the first case, the RIS-AS-TFSSK obtains about 2 dB and 1 dB gains with S $=$ 2 and S $= 4$, respectively. When N_t is increased to 8, the gains are increased to 8 dB and 5 dB under S $= 4$ and S $= 8$, respectively. The reason is that, RIS-AS-TFSSK scheme uses antenna combinations with larger channel gains to convey information, therefore better BER performance can be obtained through the channel with large gain. Although the decrease of the number of selected antenna combinations results in a reduction in spectral efficiency, more powerful combinations of channel coefficients are selected which enhances the BER performance of the system.

VI. Conclusion

In this letter, a novel TFSSK scheme based on RIS called RIS-TFSSK has been proposed in wireless communication. Theoretical derivations of BER is provided by using ML detector and simulation results are provided to evaluate the potential of the proposed scheme. In addition, an antenna selection method based on channel gain ranking is proposed and evaluated by using Monte Carlo simulations, which shows the BER superiority of RIS-AS-TFSSK scheme. The proposed can be used as a candidate for the future wireless communication systems due to its convinced BER performance and flexibility. In addition, based on the developed work, the extension into double index modulation or active RIS frameworks will be investigated in the future work.

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